

AP CSP | Chapter 5: Connected Systems Investigation

Part A | Network paths, failures and packet evidence

Name: _____ Partner: _____ Date: _____

Use the four sections of Chapter 5 Network Lab. Predict before each run. Preserve working data between observations, and reset links before each new failure set. These simulators do not measure a real Internet connection.

1. Network and resilience model

Nodes: S,A,B,C,D,T. Undirected edges: S-A, S-B, A-C, A-D, B-D, C-T, D-T, C-D. The model chooses a shortest-hop path, checking neighbors alphabetically. Record reachability, route and hops.

Disabled edges	Prediction: path / hops or no path	Observation: path / hops or no path
None		
A-C only		
S-A and S-B		

2. Packet model: split HELLO AP CSP into chunks of 3 characters. Keep both spaces. Write each payload inside quotation marks.

Packet ID	Predicted exact payload	Observed exact payload
1		
2		
3		
4		

3. Arrival order is 4,2,1,3. First lose ID 3, then retry it. Record received/missing IDs before retry and the evidence that permits complete reconstruction afterward.

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Part B | Parallel time, overhead and the chapter conclusion

4. Timing model: serial setup = 2 time units; independent indivisible jobs [6,4,3,3]. Assign jobs in listed order to the earliest free processor; ties use the lowest processor number. Start with overhead 0.

Processors	Predicted total time	Observed total time	Speedup vs 1 processor
1			
2			
4			

5. Show the two-processor schedule. Include setup and label job start/finish times.

Processor	Job sequence and times (or draw a timeline)
P1	
P2	

6. Keep the same jobs and use 2 processors. Add 6 units of overhead after the parallel phase. Predict and observe total time; calculate speedup using the original no-overhead sequential baseline.

7. Explain why more processors cannot repair a missing network path or an unreceived required packet. Cite one observation from Part A.

8. State one model limitation and one targeted improvement. Identify what your evidence supports and what it cannot guarantee.